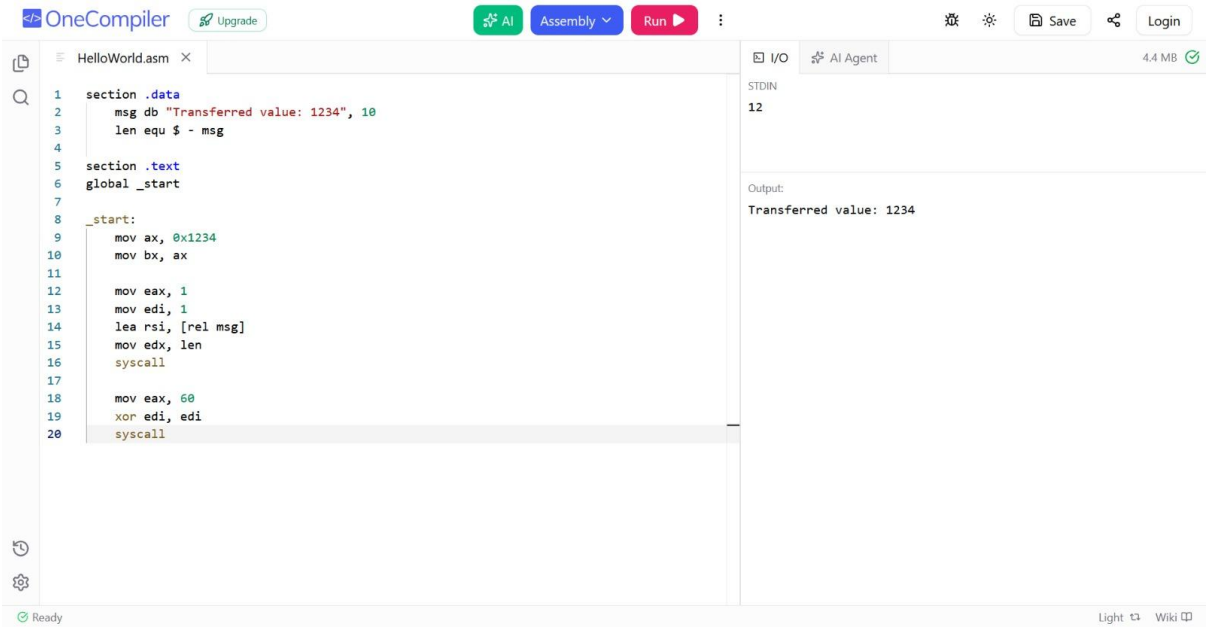
EXPERIMENT 1

B




```
section .data
    msg db "Transferred value: 1234", 10
    len equ $ - msg

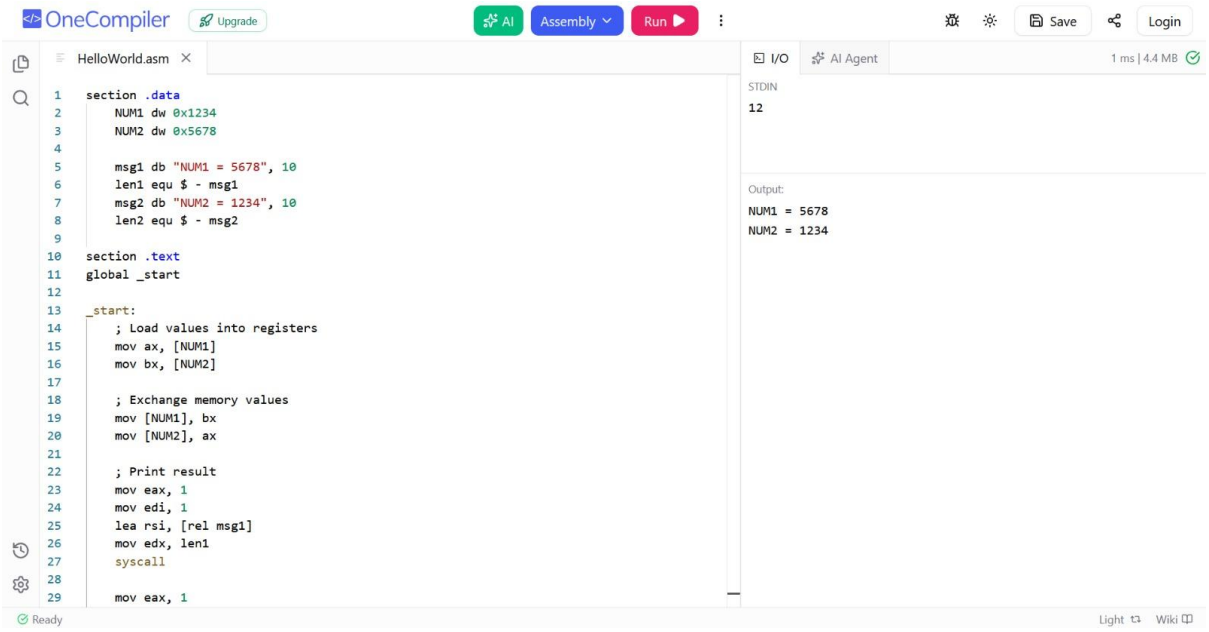
section .text
global _start

_start:
    mov ax, 0x1234
    mov bx, ax

    mov eax, 1
    mov edi, 1
    lea rsi, [rel msg]
    mov edx, len
    syscall

    mov eax, 60
    xor edi, edi
    syscall
```

C




```
section .data
    NUM1 dw 0x1234
    NUM2 dw 0x5678

    msg1 db "NUM1 = 5678", 10
    len1 equ $ - msg1
    msg2 db "NUM2 = 1234", 10
    len2 equ $ - msg2

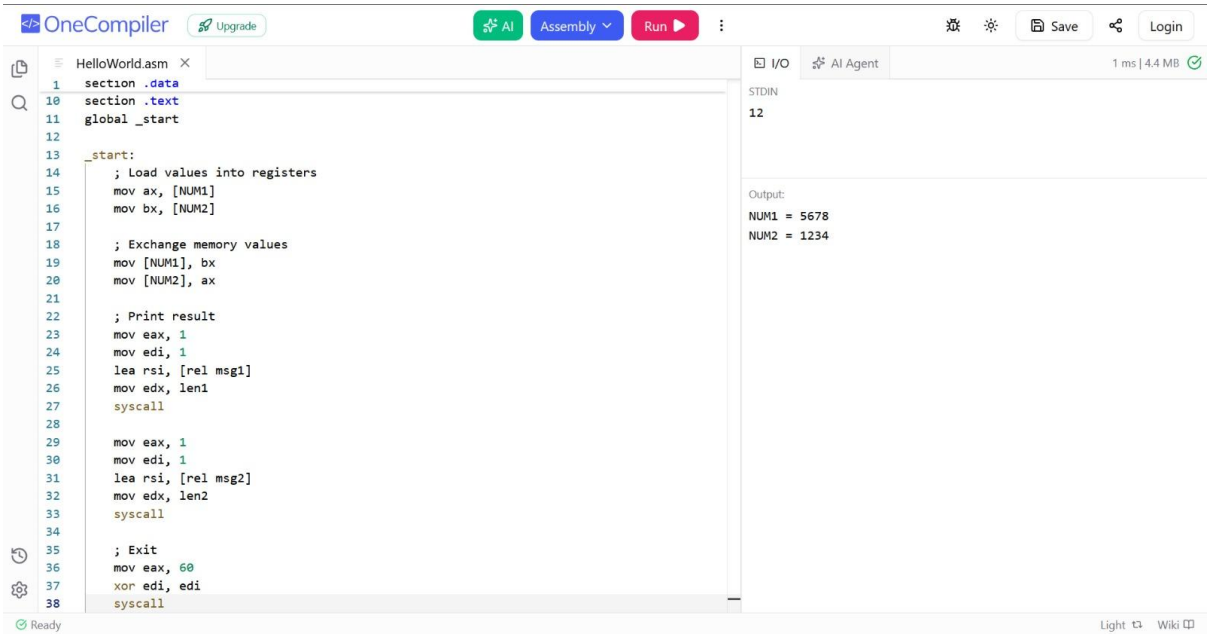
section .text
global _start

_start:
    ; Load values into registers
    mov ax, [NUM1]
    mov bx, [NUM2]

    ; Exchange memory values
    mov [NUM1], bx
    mov [NUM2], ax

    ; Print result
    mov eax, 1
    mov edi, 1
    lea rsi, [rel msg1]
    mov edx, len1
    syscall

    mov eax, 1
```


OneCompiler
Upgrade
AI
Assembly
Run
Save
Login
HelloWorld.asm
section .data
section .text
global _start
_start:
; Load values into registers
mov ax, [NUM1]
mov bx, [NUM2]
; Exchange memory values
mov [NUM1], bx
mov [NUM2], ax
; Print result
mov eax, 1
mov edi, 1
lea rsi, [rel msg1]
mov edx, len1
syscall
mov eax, 1
mov edi, 1
lea rsi, [rel msg2]
mov edx, len2
syscall
; Exit
mov eax, 60
xor edi, edi
syscall
I/O
AI Agent
1 ms | 4.4 MB
STDIN
12
Output:
NUM1 = 5678
NUM2 = 1234
Ready
Light
Wiki